CONDITIONAL STRUCTURES

GROUP 8

TABLE OF CONTENTS

- Essence of Conditional Structures
- Types of Conditional Structures
- Using Multiple Statements in If and If-Else Structures
- When to use certain type of Conditional Structures.
- Nested If and If-Else Statements in Java
- Logical AND and OR Operators in Conditional Statements

ESSENCE OF CONDITIONAL STRUCTURES

ESSENCE OF CONDITIONAL STRUCTURES

Conditional structures in programming are control flow features that allow software to make decisions. They evaluate whether a specific expression is true or false, then execute different blocks of code based on that result.

TYPES OF CONDITIONAL STRUCTURES

TYPES OF CONDITIONAL STRUCTURES

THE SIMPLE IF STATEMENT

The most basic form of decision-making. It evaluates a single boolean expression; if the condition is true, it runs the specified block of code. If the condition is false, the block is completely skipped.

THE IF-ELSE STATEMENT

This structure offers two distinct paths. It executes one block of code if the condition evaluates to true, and a completely separate alternative block (the else block) if the condition evaluates to false.

THE IF-ELSE IF (OR ELIF) LADDER

Used when you need to evaluate multiple, mutually exclusive conditions sequentially. The program checks each condition from top to bottom and executes only the first block whose condition evaluates to true. An optional catch-all else can be placed at the end.

THE NESTED IF STRUCTURE

This involves placing an if or if-else statement inside another if or else block. It is highly useful for performing multi-level or layered decision checks.

THE SWITCH (OR MATCH) STATEMENT

A cleaner, highly readable alternative to a long if-else if ladder. It takes a single variable or expression and compares it against multiple predefined constant values (called cases). In newer languages like Python, this is implemented as structural pattern matching (match-case).

THE TERNARY OPERATOR (?:)

A compact, one-liner shorthand equivalent to a basic if-else statement. It evaluates a condition and directly returns one value if true, and another if false.

USING MULTIPLE STATEMENTS IN IF AND IF-ELSE STRUCTURES

USING MULTIPLE STATEMENTS IN IF AND IF-ELSE STRUCTURES

Multiple statements are used when a condition needs to perform several actions. Curly braces {} group these statements into a block, allowing the program to execute them together.

STATEMENT

an instruction that tells the program to perform an action.

MULTIPLE STATEMENTS

two or more instructions placed on the same conditional block.

MULTIPLE STATEMENTS ARE BEST USED ON PROGRAMS LIKE:

Online Shopping System

If the condition is **true**:

1.Deduct the price.
2.Confirm the purchase.
3.Display the remaining balance.

If the condition is **false**:

1.State that it has not enough money on balance.
2.Instruct customer to add money.

Age Verification System

If the condition is **true**:

1.Confirm that the user is an adult.
2.State the user can apply for an ID.

If the condition is **false**:

1.Confirm that the user is a minor.
2.Instruct that the customer may need a guardian.

EXAMPLE CODE:

Online Shopping System

```java
// Multiple Statements
public class MultipleStatements{
    public static void main(String[] args){

        float money = 1000;
        float price = 250;

        if (money >= price) {
            money = money - price;
            System.out.println("Purchase Successful!");
            System.out.println("Remaining money: " + money);
        } else {
            System.out.println("Not enough money.");
            System.out.println("Please add more funds.");
        }
    }
}
```

```
Purchase Successful!
Remaining money: 750.0
```

EXAMPLE CODE:

Age Verification System

```java
// Multiple Statements
public class MultipleStatements{
    public static void main(String[] args){

        int age = 19;

        if (age >= 19) {
            System.out.println("You are an adult.");
            System.out.println("You can apply for an ID.");
        } else {
            System.out.println("You are a minor.");
            System.out.println("You may need a guardian.");
        }
    }
}
```

```
You are an adult.
You can apply for an ID.
```

WHEN TO USE CERTAIN TYPE OF CONDITIONAL STRUCTURES

CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

IF STATEMENT

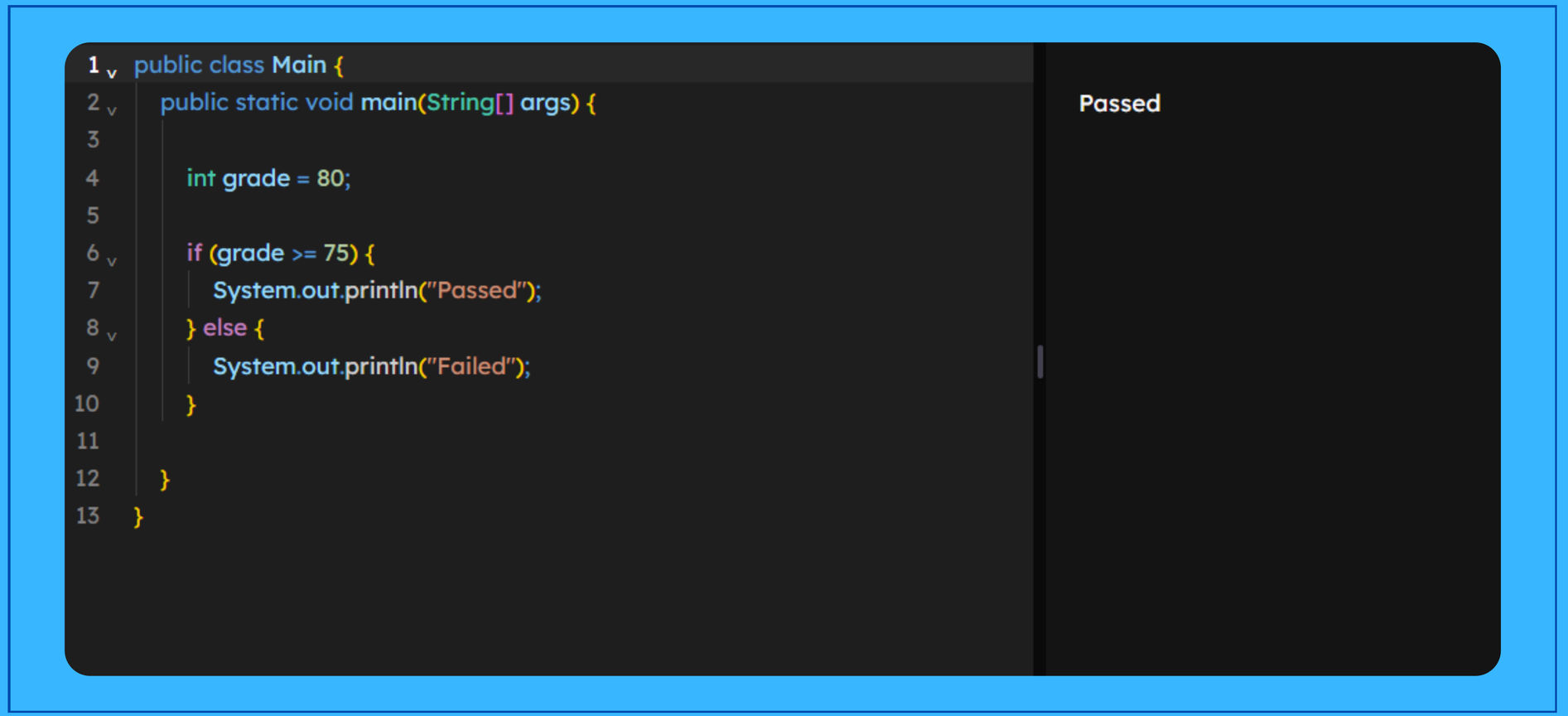
Used if only one condition is needed to check.



CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

IF-ELSE STATEMENT

Having 2 posible outcomes - one if condition is ture and one action if is false.



CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

IF-ELSE-IF LADDER

Having multiple scenarios, and it alwasys follow the If-condition.

```java
public class Main {
    public static void main(String[] args) {

        int grade = 85;

        if (grade >= 90) {
            System.out.println("Excellent");
        } else if (grade >=  80) {
            System.out.println("Very Good");
        } else if (grade >= 75) {
            System.out.println("Passed");
        } else {
            System.out.println("Failed");
        }

    }
}
```

```
Very Good
```

CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

NESTED IF STRUCTURE

Used if legitimacy or double checking is needed for a condtion.

```java
public class Main {
    public static void main(String[] args) {

        int age = 20;
        boolean hasID = true;

        if (age >= 18) {
            if (hasID) {
                System.out.println("You can enter.");
            }
        }

    }
}
```

```
You can enter.
```

CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

SWITCH CASE

Used if trying to find one value among various choices.

```java
public class Main {
    public static void main(String[] args) {
        int day = 2;

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default :
                System.out.println("Invalid day");
        }
    }
}
```

```
Tuesday
```

CHOOSING THE APPROPRIATE CONDITIONAL STRUCTURE

TERNARY OPERATOR

used for asnwering the given the simple way, technically ternary operator is a simplier version of if-else statement.

```java
public class Main {
    public static void main(String[] args) {

        int age = 20;

        String result = (age >= 18) ? "Adult" : "Minor";

        System.out.println(result);

    }
}
```

```
Adult
```

NESTED IF AND IF-ELSE STATEMENTS IN JAVA

Nested If and If-Else Statements in Java

Nested if in Java refers to placing one if statement inside another if statement. It is used when a condition depends on another condition being true before it can be evaluated. This helps in handling multiple related decisions in a structured manner.

- A nested if contains an if statement inside another if statement.

- The inner condition is checked only if the outer condition is true.

- It is useful for evaluating multiple dependent conditions.

EXAMPLES:

```java
public class Main {
    public static void main(String[] args) {

        //Outer
        if (condition1) {
            //Inner
            if (condition2) {

            }
        }

    }
}
```

```java
public class Main {
    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        if (a == 10) {
            System.out.println("MAIIPIT KA NGANI!");
            if (b == 20) {
                System.out.println("NAKAKAINIS KA EH!");
            }
        }

    }
}
```

```
MAIIPIT KA NGANI!
NAKAKAINIS KA EH!
```

EXAMPLES:

```java
public class Main {
    public static void main(String[] args) {

        //Outer
        if (condition1) {
            //Inner
            if (condition2) {

            } else {

            }
        } else {

        }

    }
}
```

```java
public class Main {
    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        if (a == 10) {
            if (b == 20) {
                System.out.println("Never Grow Old");
            } else {
                System.out.println("Chris Hippers");
            }
        } else {
            System.out.println("Kent Bryan");
        }

    }
}
```

```
Never Grow Old
```

EXAMPLES:

```java
public class Main {
    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        if (a == 10) {
            if (b == 10) {
                System.out.println("Never Grow Old");
            } else {
                System.out.println("Chris Hippers");
            }
        } else {
            System.out.println("Kent Bryan");
        }

    }
}
```

Chris Hippers

```java
public class Main {
    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        if (a == 5) {
            if (b == 20) {
                System.out.println("Never Grow Old");
            } else {
                System.out.println("Chris Hippers");
            }
        } else {
            System.out.println("Kent Bryan");
        }

    }
}
```

Kent Bryan

EXAMPLES:

```java
public class Main {
    public static void main(String[] args) {

        int studentNum = 261893;
        int birthday = 12302007;

        if (studentNum == 261893) {
            if (birthday == 12302007) {
                System.out.println("Login Successfully!");
            } else {
                System.out.println("Login Failed!");
            }
        } else {
            System.out.println("Invalid Student Number!");
        }

    }
}
```

```
Login Successfully!
```

```java
public class Main {
    public static void main(String[] args) {

        int age = 18;
        boolean isCitizen = true;

        if (age >= 18) {
            System.out.println("You are valid.");
            if (isCitizen) {
                System.out.println("You are eligible.");
            } else {
                System.out.println("You must be a citizen.");
            }
        } else {
            System.out.println("You must be 18 or older.");
        }

    }
}
```

```
You are valid.
You are eligible.
```

LOGICAL AND AND OR OPERATORS IN CONDITIONAL STATEMENTS

Logical AND and OR Operators in Conditional Statements

In Java, the logical AND (&&) and logical OR (||) operators combine two or more boolean conditions inside structures such as if, else, and else if. Use && when all conditions must be true, and || when at least one condition must be true.

LOGICAL AND

The AND (&&) operator returns true only when both conditions are true.

LOGICAL OR

The OR (|I) operator returns true when at least one condition is true. It returns false only when both conditions are false.

Logical Operator AND Examples:

```java
// Logical Operator AND
public class AND{
  public static void main(String[] args){

    boolean canDrive = true;
    boolean hasLicense = true;

      if (canDrive && hasLicense ) {
          System.out.println("Alam mo ha, pwede ka.");
      } else {
          System.out.println("Tapos kapag na disgrasya ka kakatok ka sa puso namin!? PATAWAD!!!");

    }
  }
}
```

```
Alam mo ha, pwede ka.
```

```java
// Logical Operator AND
public class AND{
  public static void main(String[] args){

    boolean canDrive = true;
    boolean hasLicense = false;

      if (canDrive && hasLicense ) {
          System.out.println("Alam mo ha, pwede ka.");
      } else {
          System.out.println("Tapos kapag na disgrasya ka kakatok ka sa puso namin!? PATAWAD!!!");

    }
  }
}
```

```
Tapos kapag na disgrasya ka kakatok ka sa puso namin!? PATAWAD!!!
```

Logical Operator OR Examples:

```java
// Logical Operator OR
public class OR{
  public static void main(String[] args){

    String day = "Saturday";

      if (day == ("Saturday") || day == ("Sunday")) {
          System.out.println("It is the weekend.");
      } else {
          System.out.println("It is a weekday.");
    }
  }
}
```

```
It is the weekend.
```

```java
// Logical Operator OR
public class OR{
  public static void main(String[] args){

    String day = "Monday";

      if (day == ("Saturday") || day == ("Sunday")) {
          System.out.println("It is the weekend.");
      } else {
          System.out.println("It is a weekday.");
    }
  }
}
```

```
It is a weekday.
```

WRITTEN ACTIVITY

I. MULTIPLE CHOICE

1. Which statement checks one condition?
A. what
B. if
C. may iba na
D. Switch

2. Which structure is useful for choosing between several specific options?
A. Ternary
B. If
C. If-else if
D. Switch

3. Which of the statements is used for many conditions?
A. If-else
B. If
C. If-else if
D. Switch

4. What are conditional structures used for?
A. Saving files
B. Playing games
C. Listening to music
D. Making decisions

5. What statement gives 2 possible outcomes?
A. If
B. If-else
C. Ternary
D. Switch

WRITTEN ACTIVITY

6. What is the purpose of default in a switch?
A. It runs when no choice matches
B. It always runs first
C. It repeats the program
D. It stops the computer

7. Which structure is a short form of if-else?
A. Switch
B. Nested if
C. Ternary
D. If-else if

8. Which conditional structure is best for a short and simple decision?
A. Ternary
B. Switch
C. Nested if
D. If-else if

9. If you need to choose between days in a week what structure is the most suitable?
A. If
B. Switch
C. Ternary
D. Nested if

10. Which logical operator means AND?
A. ||
B. &&
C. !!
D. &!

WRITTEN ACTIVITY

11. If you need to check several grade levels, which is suitable?
A. If
B. If-else if
C. Break
D. Default

12. Which structure has a true and false option?
A. If
B. If-else
C. Switch
D. Default

13. What is an if statement used for?
A. Checking a condition
B. Making a variable
C. Ending a program
D. Repeating a program

14. What happens when an if condition is false?

A. The code inside it is skipped
B. The computer shuts down
C. The code always runs
D. The program starts again

15.Which logical operator means OR?
A. ||
B. &&
C. !!
D. &!

WRITTEN ACTIVITY

II. True or False

16. Conditional structure is used for making decisions.

17. Nested if in Java refers to placing one if statement inside another if statement

18. If statement is used if two outcomes is needed.

19. If-else if ladder is used for multiple conditions.

20. If-else statement is used for multiple coniditions.

21. Switch case is used for finding one value among other choices.

22. Ternary is the simplier version of If-else statement.

23. A statement is an instruction that tells the program to perform an action.

24. In Java, the logical AND (&&) and logical OR (||) operators combine two or more boolean conditions inside structures such as if, else if, while, and for.

WRITTEN ACTIVITY

III. Enumeration

25-30 : Give the six conditional structures in any order.

References:

https://www.scribd.com/document/977790151/4c3b484411de9a55ef3cc5d2056799c0645753-29329-691ec78172ee8
https://www.scribd.com/presentation/491257667/conditional-control-structure-chap-04-class-10
https://www.allpcb.com/allelectrohub/using-conditional-structures-in-c
https://https://dev.to/ricardocaselati/exploring-conditional-structures-in-java-3f2jwww.youtube.com/watch?v=MY03bt_0JQI
https:/https://www.geeksforgeeks.org/java/decision-making-javaif-else-switch-break-continue-jump//java-programming.mooc.fi/part-1/6-conditional-statements
https://chatgpt.com/c/6ab4722c-8188-83ec-b67c-2360a9c34912
https://www.geeksforgeeks.org/java/nested-if-in-java/
https://www.w3schools.com/java/java_conditions_logical.asp

THANK YOU

THANK YOU
THANK YOU